

RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ??????????? ?????????????? ????????? ? ??????? ???????
> A/B ? CUPAC A ? A+B ? A+B+C; ????????? ?????????? open-source ?????????? ? ?????????? ???.
> ????????? ??????? ?? ??????? ? notebook.ipynb (??????????, seed=11).

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?????????, ??? ? ? ?????????????????? ?????????? ?????????? in-time ?????????? ??????
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ????????????? ?????????? ?????????) ???
???????????? ?????????????????? ?????????????? ??????? ATE.

?????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;
? **B** expert-safe in-time ? ????????? second-stage ??????????
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;
? **D** dag-required ? ??? estimator (????? causal-????????);
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ????? ATE;
? **unsafe_demo** ? ?????? ?????????????? ??????, ?? ?????????.

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ab_hypex (A/B baseline) • hypex_cupac (reference/parity, ??? ??????) •
sklearn_cupac_A ? sklearn_cupac_A_plus_B_linear ? sklearn_cupac_A_plus_B_plus_C_linear
(????????? ?????????? ??????????) • unsafe_demo_optional (????????????).

???????????? ? ? ?????????? (N=8000, SEED=11, ????????? ATE = 2.8)

method	ate	p_value	variance_reduction_vs_ab_%	incremental_vs_predecessor_%	safety_status
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ab_hypex	2.758	0.0	0.0	?	ok
hypex_cupac	2.785	0.0	13.56	?	reference_only
sklearn_cupac_A	2.801	0.0	25.63	25.63	ok
sklearn_cupac_A_plus_B_linear	2.775	0.0	42.38	22.52	ok
sklearn_cupac_A_plus_B_plus_C_linear	2.766	0.0	52.80	18.07	ok
unsafe_demo_optional	1.000	0.0	83.17	?	unsafe_demo

(????????? ?????????? ????????? ? ??????? ?????????????? ?????????? ?? ?????????? ??????????)

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1. ?????? ?????????????? ??????? ???? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)
????????? ?????????? ?????????????? A/B, ??? ???? ATE ?????????? ?????????????? (? 2.8).
2. CUPAC parity: hypex_cupac (~14%) ?????????? ?? ??????? ? sklearn_cupac_A;
????????????? ?????????????? ????????? ???? R&D ? ?????????????????? ??? ?????????.
3. Balance/missingness gate ?????????????? A/B/C (|SMD|?0) ? ?????????? D/E/F ? unsafe_demo
(|SMD| ? 0.45?1.14, p?0) ? ?????????, ??????? ?? ??????? ?????? ? estimator.
4. unsafe_demo ???? ?????????????????? «????????? ??????????» (~83%), ?? ?????????? ??????? ATE

?????? ?? ????????? 2.8 ? ????????? ? 1.0 (??, ???????) ? ?????????????? ???????
outcome-derived ??????????; ? ?????????????? ??????? ?? ???????.

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? balance-test ? ????????????? gate (????? ????????????? ?????????? E/F), ? **??** ?????????????????
???????????? ?????????????????; ??? D-???????????? ?????? ????????? DAG/causal-?????
? E/F ?????????????; safe-intime ?????????????? **????????** ? ?????????? second-stage, ?? ?
???????????????? ML-????????.

???????????????? ?? ????????? ? HYPEX

?????? ????????????? ????????????? ????????? ? ????????? ? ????????? ? HypEx CUPAC. ?????
?????????
? HypEx ??????????: (1) ????????????? ?????? B/C ?? ????????????? ?????????? open-source ??????????;
(2) ????????????????? balance/missingness gate ??? ?????????????????????? ??????????; (3) ?????????????????
???????? E/F ? ????????????? ?????? unsafe_demo. ????????????? ?????????????????? ????????????????? CUPAC ??
???????????? ?????? ? VarWar (???????????? ????????????? ~34% ? ~33%; ??????? ??????????, ?? ?????????????).

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bash
pip install -e .[ml]
??. NOTEBOOK.IPYNB; ?????? ATE–CI ? ?????????? ?????????????? ?? RND_REPORTS.BENCHMARK

python tools/generate_pdf.py # ?????????????? report.pdf